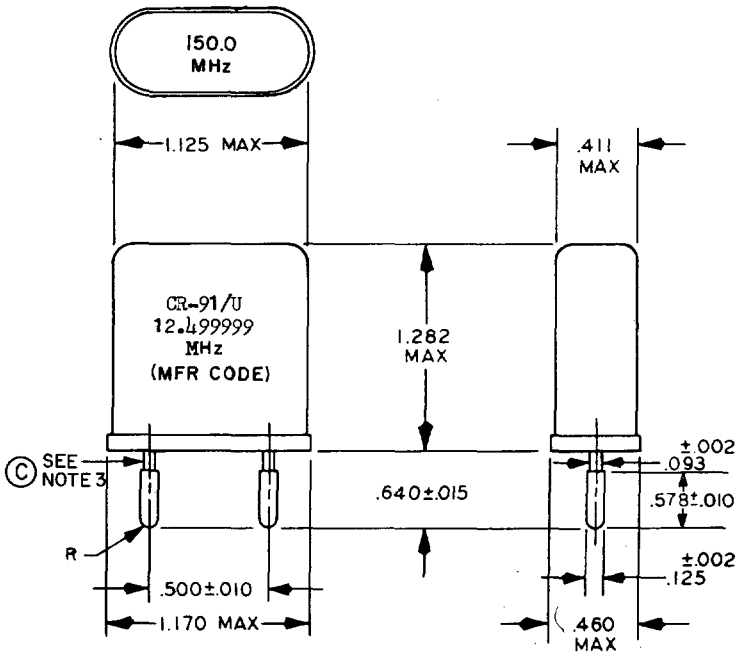
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, TYPE CR-91/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the crystal unit described herein shall consist of this document and the latest issue of Specification MIL-C-3098.

Pertinent characteristics: 9.5 to 12.5 MHz; fundamental; noncontrolled; antiresonance.



INCHES	MM	INCHES	MM
.002	.05	.500	12.70
.010	.25	.578	14.68
.015	.38	.640	16.26
.093	2.36	1.125	28.58
.125	3.18	1.170	29.72
.411	10.44	1.282	32.56
.460	11.68		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
© 3. The pin undercut may be omitted.

© denotes change

FSC 5955

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TABLE I. Requirements.

Requirement	Specified value or condition
Holder	HC-11A/U (MIL-H-10056/17)
Dimensions, marking, and configuration	See figure
Frequency range	9.5 to 12.5 MHz, inclusive
Capacitance, shunt	10 pF, maximum
Frequency tolerance (operating temperature range)	±50 PPM
Equivalent resistance	25 ohms, maximum
Antiresonance, load capacitance	55.0 ± 0.5 pF
Mode of oscillation	Fundamental
Operating temperature range	-55° to +90°C, inclusive
Government reference standard test set	TS-330/TSM
Calibration values:	
Resistance	12 ohms
Crystal current	45 mA
Drive level, rated	25 mW
Shock (specified pulse):	
Frequency change permitted	±5 PPM
Equivalent resistance change permitted	±10 percent
Vibration: 1/	
Frequency change permitted	±5 PPM
Equivalent resistance change permitted	±10 percent
Temperature run:	
Frequency change permitted	±5 PPM
Equivalent resistance change permitted	±10 percent
Aging:	
Frequency change permitted	5 PPM

1/ Method 201 of MIL-STD-202.

NOTES:
1. Marking. The operating frequency shall be marked on the top of the crystal unit and the fundamental frequency on the side of the crystal unit as shown in figure 1.
2. Intended use. These crystal units have previously been misidentified as "CR-1A/AR." Type CR-91/U crystal units are for use in Aircraft Radio Corporation type 12 equipment.
3. In using equipment, these frequencies are multiplied to the 12th harmonic which is also known as air frequency.
4. This crystal-unit type is to be used for special applications and for replacement purposes only. Specific approval of the Government is required prior to each intended use (see MIL-STD-683).

Custodians:
Army - EL
Navy - EC
Air Force - 84

Review activities:
Army - EL, MI, MU, SG, SM
Navy - EC
Air Force - 11, 85
DSA - ES

User activities:
Army -
Navy - AS, MC, SH
Air Force - 17, 19

Preparing activity:
Army - EL

Agent:
DSA - ES

(Project 5955-0408-38)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-C-3098/68C CRYSTAL UNIT, QUARTZ, TYPE CR-91/U

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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Department of the Army
Headquarters
U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

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